

TianShu - a half Humanoid Robotic System with Mecanum Wheel Base

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Abstract

This paper introduces a Mecanum wheel humanoid robot designed for industrial automation that specializes in object recognition, assembly, and defect detection. Featuring two 7-DOF robotic arms with independently controlled grippers, the system achieves high precision and adaptability in dynamic environments.

The robot's omnidirectional mobility enables smooth navigation in confined spaces, while stepper motors with high-reduction harmonic gearboxes (50:1) provide exceptional torque and accuracy. A closed-loop control system with encoders ensures precise movement, and integrated force and weight sensors allow for safe handling of fragile objects and defect detection. Additionally, the Nicla Vision module with YOLO 11-OBb enables real-time AI-powered object recognition, improving its automation capabilities. A built-in real-time display improves human-robot interaction, making operation more intuitive.

Keywords: *Mecanum-wheel; humanoid robot; 7-DOF robotic arm; YOLO Object Detection*

1 Introduction

Nowadays, an increasing number of industrial production line tasks that require high precision and extreme repetition have become too challenging for humans to handle. As a result, attention has shifted toward robots, which offer superior accuracy and never experience fatigue. A significant portion of low-skilled labor has been replaced by robotic arms or humanoid robots. Consequently, many companies have achieved a 30% increase in productivity, a 50% reduction in production costs, and a utilization rate exceeding 85[1].

Our system is a humanoid robot based on Mecanum wheels. It is equipped with two robotic arms, each having seven degrees of freedom, and both arms feature independently controlled grippers. The robot is designed for industrial production lines, where it can perform tasks such as object recognition, assembly, and defect detection with high flexibility and precision. The use of Mecanum wheels allows for omnidirectional movement, which enables the robot to navigate efficiently in constrained or dynamic environments.

The system is designed with extremely high operational precision. It utilizes stepper motors combined with a high-reduction harmonic gearbox (50:1), which not only enhances precision but also increases torque. For motion accuracy, the system employs DC motors with encoders, ensuring precise positioning and movement. It offers excellent maneuverability, featuring humanoid arms with seven degrees of freedom, similar to a human's natural movements. Regarding mobility, the robot is equipped with four Mecanum wheels, enabling agile movement in confined spaces. A load scale sensor is integrated to measure object weight, allowing for precise differentiation between components based on subtle weight variations and enabling defect detection. The system can also monitor gripping force, making it capable of handling fragile objects without causing damage. In addition, it can detect accidental drops of objects. A built-in screen

displays operational data in real time, enhancing monitoring, improving human-machine interaction, and showing AI-related information for better user engagement. The gripper is equipped with a Nicla Vision module, combined with YOLO 11 Oriented Object Detection and the high AI computational power of the Raspberry Pi 4, enabling real-time, low-latency detection of the relative position and orientation between the gripper and the target object.

Tesla unveiled its humanoid robot project, "Optimus," at the AI Day event in August 2021. The goal of the project is to automate dangerous, repetitive, and mundane tasks. Optimus is a bipedal robot with a maximum speed of 8 kilometers per hour. It has a human-like build, weighing around 57 kilograms and standing 1.73 meters tall[2]. Tesla's proprietary technologies, including autonomous driving cameras, neural network planning, and simulation capabilities, will be integrated into the robot.

Honda Motor Co., Ltd. began developing a mobile humanoid robot named "Advanced Step in Innovative Mobility" (ASIMO) in 1986. The first version of ASIMO was launched in 2000, showcasing two key human-like features: bipedal walking and the ability to "understand and learn from humans." In addition to maintaining stable walking on uneven and inclined surfaces, ASIMO is equipped with various other features[3].

In comparison, our robot has several advantages. It is more cost-effective and consumes less power. Additionally, the screen on its head is better suited for human-robot interaction, enhancing communication and ease of use. Our robot also performs better on complex, uneven terrains, allowing it to work in a wider range of environments. Furthermore, our robot has the ability to perform rotational target detection, a long lifespan, and a lower cost.



Figure 1: TianShu

2 Theoretical Background

2.1 Schematic Diagram and Flow Chart

Below is a schematic diagram of the system with clear labels of all parameters and variables/symbols.

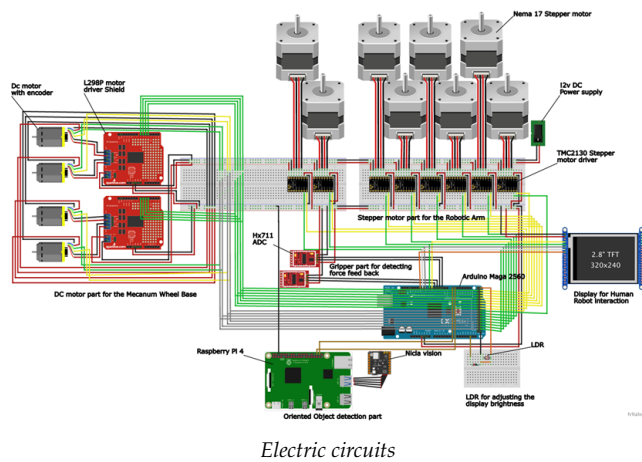


Figure 2

The system operates as an intelligent robotic manipulator that integrates vision-based object detection, inverse kinematics for motion planning, and precise motor control for object grasping and transportation. The Nicla Vision Module as shown in figure 7a captures a live video stream and transmits it to the Raspberry Pi via serial port, which runs a YOLO 11-OBB based object detection algorithm to determine the position and orientation of the target object as shown in the flowchart in 7. Using inverse kinematics, the Raspberry Pi computes the required joint angles for positioning the robotic arm and transmits these values to the Arduino Mega 2560 via I2C protocol, which serves as the motion controller. The Arduino then generates PWM signals to control the TMC2230 stepper motor drivers via the dir and step pins, which drive the stepper motors, moving the robotic arm precisely to the target location.

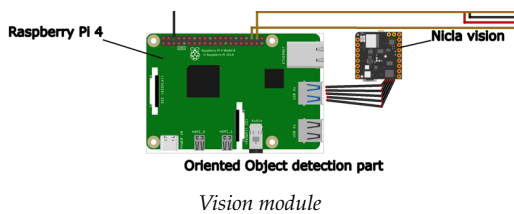


Figure 3

Once the claw reaches the object, the Arduino sends control signals to close the gripper, while a force sensor, processed through an HX711 ADC module, detects contact and stops the closing action once sufficient force is applied, ensuring a secure but non-damaging grip. The claw can then rotate 90 degrees to measure the object's weight using the force sensor before the IK solver computes the required joint angles for transporting and placing the object at the designated location. If the target position is beyond the manipulator's reach, the Raspberry Pi calculates the necessary displacement and orientation adjustments and sends these values to the Arduino, which computes the required motor angles.

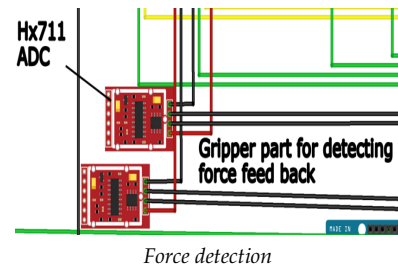


Figure 4

The Arduino then sends signals to the L298P motor driver shields, which control the four DC motors responsible for repositioning the robot before resuming the grasping operation.

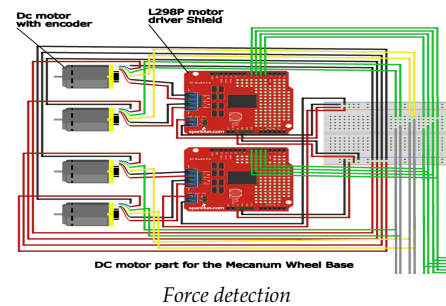


Figure 5

Additionally, the robot features a display on its head, controlled by the Raspberry Pi, which provides visual feedback using emojis, while an LDR sensor detects surrounding brightness, allowing the system to dynamically adjust the screen brightness for optimal visibility. This integrated workflow ensures efficient object detection, manipulation, and mobility, making the system highly adaptive to various real-world applications.

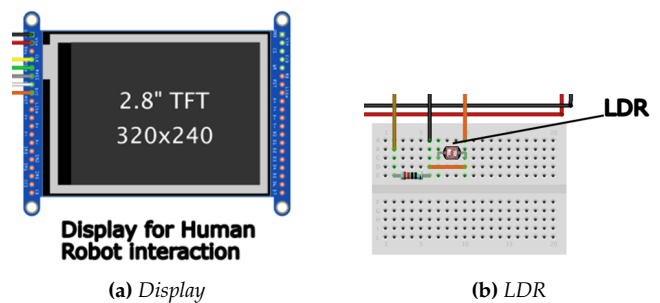
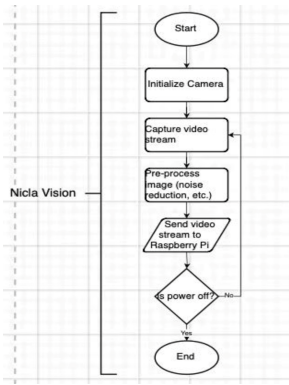
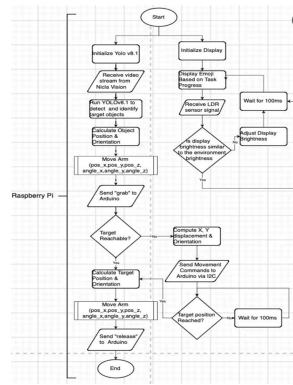


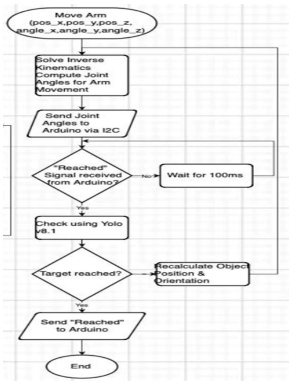
Figure 6



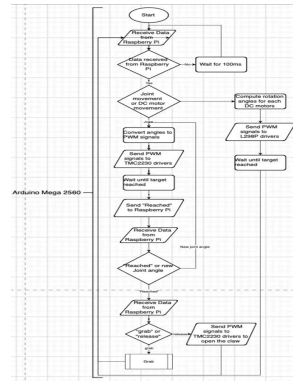
(a) Nicla Vision



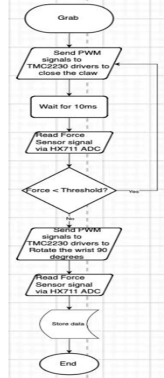
(b) Raspberry Pi Main



(c) Raspberry Pi Subroutine



(d) Arduino Mega Main



(e) Arduino Mega Subroutine

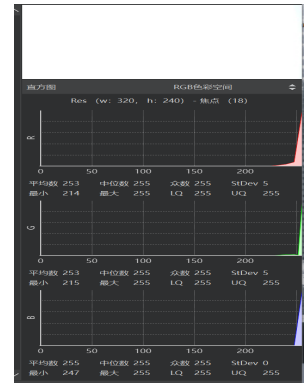
Figure 7

2.2

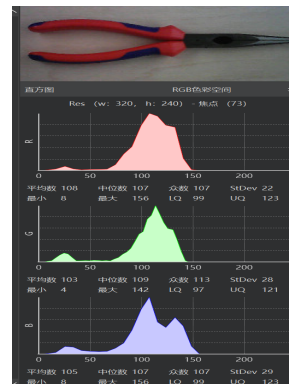
As shown in Figure 8, the first image is entirely black, with all RGB values at 0, showing absolute darkness and no color variation. The second image is completely white, with all RGB values maxed at 255, indicating uniform brightness and no contrast. The third image shows a plier with red and blue handles on a light table, with dominant red and blue peaks in the histogram. The fourth image captures our three team members with a well-balanced color distribution, showing broad RGB histograms with high variance.



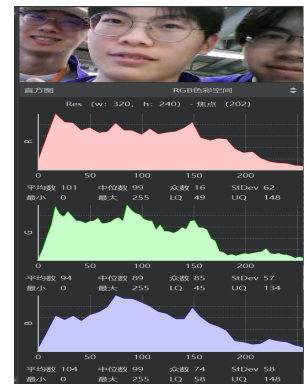
(a) RGB-1



(b) RGB-2



(c) RGB-3



(d) RGB-4

Figure 8: Images RGB-1, RGB-2, RGB-3, and RGB-4

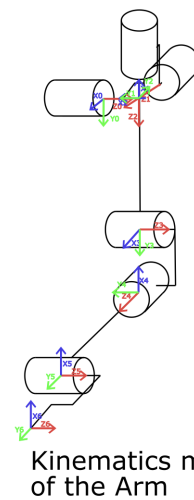


Figure 9: 6 DoF Robotic Arm Axis

2.3 Robotic Arm Motion Model

For the 7 DoF robotic arm, we decided to separate the final wrist link for additional orientation control linked with oriented object detection. For the rest 6 joint, we constructed the DH parameters as shown in table 1 using the Robotic system we designed as shown in figure 9 [5].

Joint i	d_i (mm)	a_i (mm)	α_i (rad)	θ_i (variable)
1	150	0	$\frac{\pi}{2}$	θ_1
2	0	0	$\frac{\pi}{2}$	θ_2
3	313.2	0	$\frac{\pi}{2}$	θ_3
4	0	80.5	$\frac{\pi}{2}$	θ_4
5	268.2	0	$\frac{\pi}{2}$	θ_5
6	0	92.675	0	θ_6

Table 1: Denavit-Hartenberg Parameters of the 6-DOF Robotic Arm

We then used the DH parameters to construct the Forward Kinematics Matrix [4]

$$\begin{aligned}
 T_1 &= \begin{bmatrix} \cos \theta_1 & 0 & \sin \theta_1 & 0 \\ \sin \theta_1 & 0 & -\cos \theta_1 & 0 \\ 0 & 1 & 0 & 150 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 T_2 &= \begin{bmatrix} \cos \theta_2 & 0 & \sin \theta_2 & 0 \\ \sin \theta_2 & 0 & -\cos \theta_2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 T_3 &= \begin{bmatrix} \cos \theta_3 & 0 & \sin \theta_3 & 0 \\ \sin \theta_3 & 0 & -\cos \theta_3 & 0 \\ 0 & 1 & 0 & \frac{1566}{5} \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 T_4 &= \begin{bmatrix} \cos \theta_4 & 0 & \sin \theta_4 & \frac{161 \cos \theta_4}{2} \\ \sin \theta_4 & 0 & -\cos \theta_4 & \frac{161 \sin \theta_4}{2} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 T_5 &= \begin{bmatrix} \cos \theta_5 & 0 & \sin \theta_5 & 0 \\ \sin \theta_5 & 0 & -\cos \theta_5 & 0 \\ 0 & 1 & 0 & \frac{1341}{5} \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 T_6 &= \begin{bmatrix} \cos \theta_6 & -\sin \theta_6 & 0 & \frac{3707 \cos \theta_6}{40} \\ \sin \theta_6 & \cos \theta_6 & 0 & \frac{3707 \sin \theta_6}{40} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

$$T = T_1 \cdot T_2 \cdot T_3 \cdot T_4 \cdot T_5 \cdot T_6$$

We can then use the analytical approach to find the invers kinematic model using Matlab.

2.4 Mecanum Wheel Motion Model

Assume the following definitions:

- v_x : velocity along the x-axis (lateral movement)
- v_y : velocity along the y-axis (forward and backward movement)
- ω_z : angular velocity about the z-axis (clockwise or counterclockwise rotation)

The motion model of the wheels is related to the input velocities as follows:

$$\begin{bmatrix} \omega_{FL} \\ \omega_{FR} \\ \omega_{BL} \\ \omega_{BR} \end{bmatrix} = \frac{1}{r} \begin{bmatrix} 1 & -1 & -L \\ 1 & 1 & L \\ 1 & 1 & -L \\ 1 & -1 & L \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ \omega_z \end{bmatrix}$$

where:

- $\omega_{FL}, \omega_{FR}, \omega_{BL}, \omega_{BR}$ are the angular velocities of the four motors.
- r is the radius of the wheels.
- L is the distance from the Mecanum wheels to the center of the vehicle.

2.4.1 Target Position and Conversion to Radians

Assuming the initial position of the four wheels is 0, the target position is given by:

$$x = 5m, \quad y = 6m, \quad \text{rotation about the z-axis } z = 30^\circ$$

Converting to radians:

$$\omega_z = 30^\circ = 30^\circ \times \frac{\pi}{180} = \frac{\pi}{6} \text{ radians}$$

2.4.2 Total Travel Distance

Using the Pythagorean theorem, the total travel distance d of the vehicle is:

$$d = \sqrt{x^2 + y^2} = \sqrt{5^2 + 6^2} = \sqrt{25 + 36} = \sqrt{61} \approx 7.81m$$

2.4.3 Velocity and Time Calculation

Assuming the total velocity:

$$v_{\text{total}} = 1m/s$$

The total time required to reach the target:

$$t = \frac{d}{v_{\text{total}}} = \frac{7.81}{1} = 7.81 \text{ seconds}$$

$$v_x = \frac{x}{t} = \frac{5}{7.81} \approx 0.64 \text{ m/s}$$

$$v_y = \frac{y}{t} = \frac{6}{7.81} \approx 0.77 \text{ m/s}$$

2.4.4 Wheel Rotation Calculation

Given that the wheel diameter is 96 mm, we can compute the wheel radius:

$$r = \frac{96}{2} = 48 \text{ mm} = 0.048 \text{ m}$$

For each wheel:

$$\omega_{FL} = \frac{1}{0.048} (0.64 - 0.77 - 0.3 \times 0.523) \approx -5.98 \text{ rad/s}$$

$$\omega_{FR} = \frac{1}{0.048} (0.64 + 0.77 + 0.3 \times 0.523) \approx 32.65 \text{ rad/s}$$

$$\omega_{BL} = \frac{1}{0.048} (0.64 + 0.77 - 0.3 \times 0.523) \approx 26.11 \text{ rad/s}$$

$$\omega_{BR} = \frac{1}{0.048} (0.64 - 0.77 + 0.3 \times 0.523) \approx 0.56 \text{ rad/s}$$

The rotation angles for each motor over the time period $t = 7.81s$ are calculated as:

$$\theta_i = \omega_i \times t$$

where ω_i is the angular velocity of each wheel.

Finally, calculating the rotation angles for each wheel respectively:

$$\theta_{FL} = -2.99 \cdot 7.81 \approx -23.36 \text{ radians}$$

$$\theta_{FR} = 16.32 \cdot 7.81 \approx 127.41 \text{ radians}$$

$$\theta_{BL} = 13.06 \cdot 7.81 \approx 102.02 \text{ radians}$$

$$\theta_{BR} = 0.28 \cdot 7.81 \approx 2.19 \text{ radians}$$

2.5 Input and Output signal of HX711

Excitation (+): Red Excitation (-): Black Signal (+): Green Signal (-): White Cable: 4-color wire, 0.6mm diameter, 110mm length.

Input a sine wave signal, and through PWM technology, convert the sine wave into the corresponding PWM digital signal.

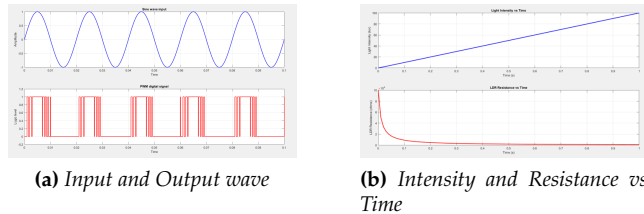


Figure 10

Simulate a gradually increasing light intensity input and calculate the resistance value of the LDR based on the light intensity. The trends of light intensity and LDR resistance over time are displayed separately.

3 Electronics Composition

3.1 Introduction of Nicla Vision

The Arduino Nicla Vision is a ready-to-use, standalone camera for analyzing and processing images on the edge. With its 2 MP color camera, smart 6-axis motion sensor, integrated microphone and distance sensor, it is suitable for asset tracking, object recognition and predictive maintenance. Quickly implement sensor nodes to send collected data via integrated Wi-Fi/Bluetooth LE connectivity. Diagrams below are the top view and back view of nicla vision with their ref.and description.

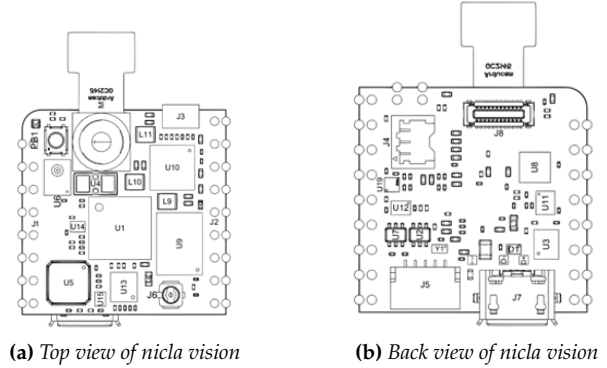


Figure 11

Ref.	Description
U1	STM32H747AI16 Dual-core Arm Cortex-M7 M4 microcontroller
U5	USB3320C-EZK-TR Hi-Speed USB 2.0 ULPI Transceiver
U9	LBEE5KL1DX-883 Type 1DX Wi-Fi + Bluetooth module w/ Infineon CYW4343W
U13	AT25QL128A-UUE-T 128 Mbit Quad SPI flash memory
U15	DSC6151HI2B-027.0000T 27 MHz Ultra-Low Power MEMS Oscillator
L9,L10,L11	DFE201610P-1R0M=P2 1 H fixed surface-mount inductor
J6	U.FL-R-SMT-1(60) Male micro UFL surface-mount coaxial connector

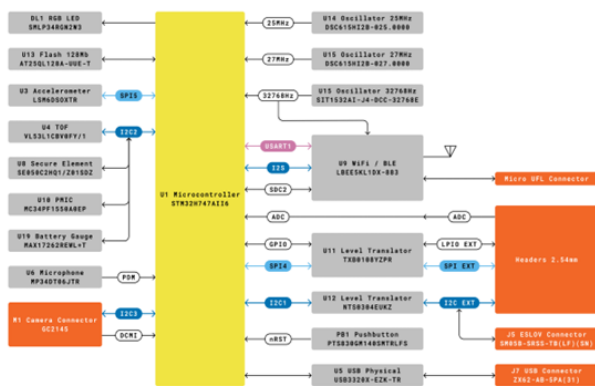
Table 2: Component Table - Part 1

Ref.	Description
M1	GC2145 2 Megapixel CMOS Camera
J1	ADC, SPI and LPIO pin headers
J3	Battery connection pin headers
U2, U7	LM66100DCKR Low-IQ Ideal Diode 5.5 V, 1.5 A, 79m
U8	SE050C2HQ1/Z01SDZ IoT Secure Element IC
U12	NTS0304EUKZ 4-bit dual-supply translating transceiver with open-drain
L3	MLF1608C330KTD00 33 H shielded multilayer inductor, 2 mA, 2.2 Ohm max, 0603 surface-mount

Table 3: Component Table - Part 2

The Nicla Vision block diagram visually represents its hardware architecture and data flow, helping understand key

components like the STM32H747 processor, GC2145 camera, wireless communication (BLE/Wi-Fi), storage, and power management. It illustrates how data moves between modules, aiding in development, debugging, and optimization.



Features: 24-bit ADC: High resolution for precise measurements. Two differential inputs: Supports two load cell channels. Selectable gain: Programmable gain of 32, 64, or 128. Low cost: Budget-friendly for weight scales or force measurement projects. Low power consumption: Operates at 1.5mA, making it suitable for battery-powered applications.

Once connected to the load cell, the HX711 outputs two key signals: DAT: Data pin, connected to a microcontroller's GPIO pin. CLK: Clock pin, connected to another GPIO pin.

Pinout and Wiring:

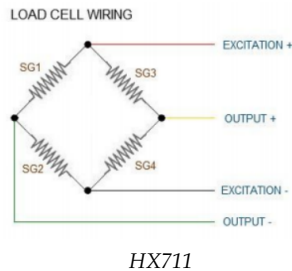


Figure 15

The HX711 has five main wires for connecting to a load cell: RED (E+): Excitation+ (VCC). BLACK (E-): Excitation- (ground). WHITE (O+): Output+ (Signal+). GREEN (O-): Output- (Signal-). YELLOW (optional EMI shielding): Used for grounding or electromagnetic interference (EMI) protection.

3.3 Photoresistors

The GL5516 series CdS photoresistors are semiconductor devices whose resistance changes with light intensity. This series is widely used in applications such as toys, lighting, camera auto exposure, light control, indoor light adjustment, industrial control, and electronic toys due to its compact size, fast response, and high sensitivity.

Features: Encapsulation and Reliability: Epoxy resin encapsulation ensures good weather resistance and reliability. Compact Size: Suitable for applications with limited space. High Sensitivity and Fast Response: Provides a significant change in resistance under low and high light conditions with fast response times. Good Spectral Response: Peak wavelength response is around 540nm, making it adaptable to various light environments.

Applications: Camera auto exposure systems Light control and indoor light adjustment Industrial control systems Light-controlled switches and lighting Electronic toys

3.4 Load cell

The TAL221 is a miniature load cell designed for precise weight measurement applications. It is ideal for use in body scales, hand scales, kitchen scales, postal scales, fishing scales, baby body scales, and other micro-electronic weighing systems. The load cell is constructed from aluminum alloy and features a parallel beam type design.

Features: Capacity: Available in ranges of 100g, 150g, 200g, 300g, 500g, 750g, 1000g, and 1500g. Material: Made from aluminum alloy. Type: Parallel beam load cell. Protection: IP65

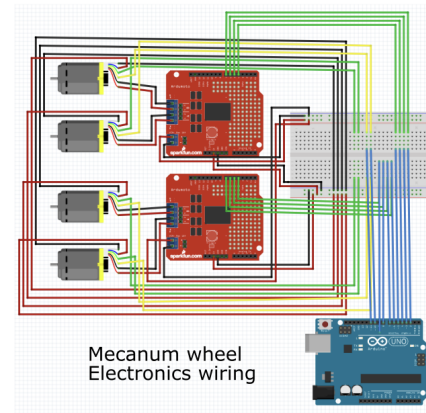


Figure 16: Mecanum Wheel

rating for protection against dust and water. Applications: Suitable for body scales, kitchen scales, postal scales, fishing scales, baby scales, and other small electronic weighing systems.

4 Electrical and Electronics Design

For our final prototype, we made several changes to our original hardware choices and communication methods.

4.1 Base

Firstly, we decided to use a separate Arduino UNO board to control the Mecanum wheels of the robot. as shown below. This adjustment minimizes entanglement and provides greater flexibility during debugging. The Arduino UNO sends PWM signals to the STEP and DIR pins on the motor shield for each motor, allowing us to control the direction and speed of motor rotation. The actual velocity and direction of rotation are then sent by the Hall sensor of each motor to Arduino via PWM signals as shown in Figure 16. The A and B outputs are square waves from 0 V to Vcc approximately 90° out of phase. The speed of the motor can be determined from the frequency, and the direction of rotation can be determined from the order of the transitions.[6] Through the speed and direction feed back, a PID close loop control has been achieved to move the robot more accurately.

4.2 Nicla Vision

For the Nicla Vision Sensor used in Oriented Object Detection, we utilized a micro USB connection between it and the

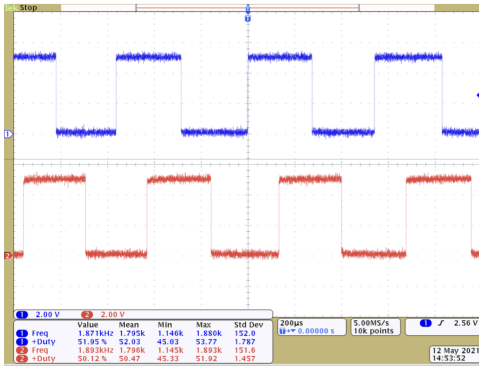


Figure 17: Hall Sensor Plot

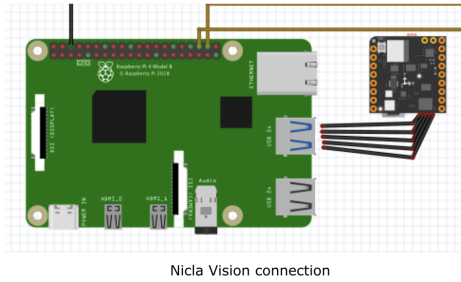


Figure 18: Nicla Vision

Raspberry Pi. This choice ensures high-speed communication while also reducing entanglement, as shown in Figure 18.

4.3 HX711

The communication between the Arduino and HX711 utilized the I2C protocol, where the analog signals from the two strain gauges in the load cell are amplified and converted into digital signals for accurate measurement. The HX711, a 24-bit analog-to-digital converter specifically designed for weighing applications, ensures high resolution and minimal noise interference. The gain of the amplifier is typically set to 128, allowing for precise weight measurements. The processed digital data is then transmitted to the Arduino for further computation and control, as shown in Figure 19.

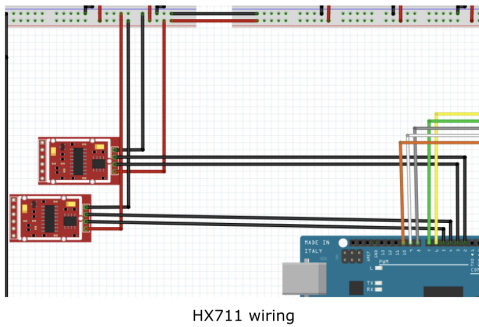


Figure 19: HX711

4.4 ILI9341

To control the ILI9341 TFT Display, we used SPI protocol which offers several advantages, including high-speed data transfer, crucial for smooth graphics rendering, enabling simultaneous data transmission and reception. It requires only four main lines (MOSI, MISO, SCLK, and CS), reducing wiring complexity compared to parallel interfaces. SPI is highly efficient, leveraging hardware SPI modules and DMA support in microcontrollers to offload processing. It provides flexible clock speed adjustment, allowing optimization for different display resolutions while consuming less power than parallel interfaces, making it ideal for battery-powered devices.

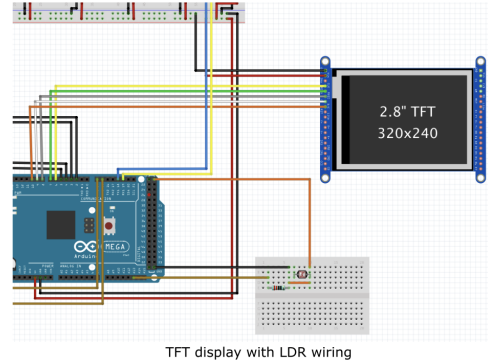


Figure 20

4.5 LDR

To adjust the brightness of the display, we utilized an Light Dependent Resistor and implemented the wiring method outlined as shown in Figure 20. This method created a voltage divider circuit, with the voltage between the LDR being detected using an analog pin on the Arduino. The changing voltage reflects variations in the LDR's resistance, which indicates the brightness of the surrounding environment. The Arduino processes this brightness information and calculates the appropriate PWM signal to send to the Lite pin on the TFT display, allowing it to adjust its brightness accordingly. The relationship between the Brightness and resistance is shown in Figure 21[6].

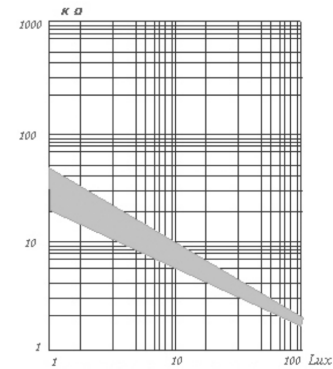


Figure 21

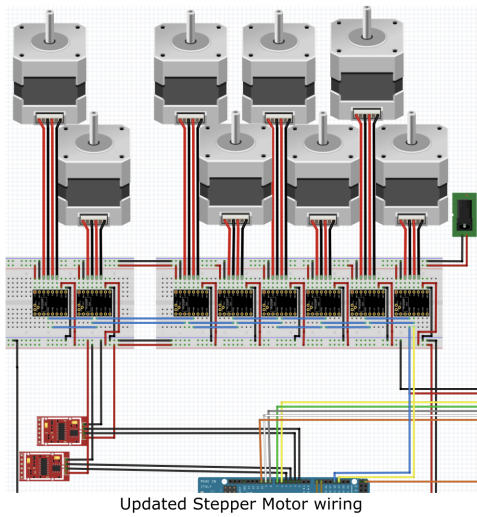


Figure 22

4.6 Stepper Motor

To effectively control the eight stepper motors on the robotic arm, we enhanced our communication approach and motor driver. We moved away from using the Arduino to send PWM signals directly to the STEP and DIR pins of the TMC2230 motor driver for open-loop control. Instead, we adopted the UART protocol for seamless communication between the Arduino and the MKS SERVO42C motor driver, ensuring improved performance and close loop control. This method provide higher flexibility on controlling the stepper motors as well as improving debugging. By sending different command, a better control over these stepper motors can be achieved, here is an example.

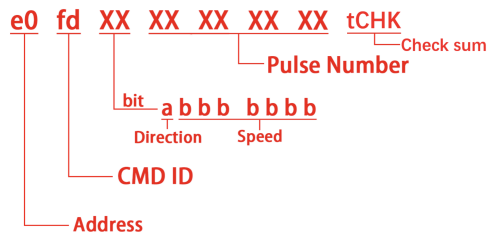


Figure 23: Command

4.7 Important Mechanical Sub-system

There are two important mechanical sub-system in this project:

4.7.1 Wrist

The robotic wrist employs three 90-degree bevel gears to achieve two degrees of freedom: pitch and yaw as shown in Figure 24. This design strategically positions the two heavy stepper motors closer to the joint, effectively reducing the moment and improving overall stability. The system utilizes two GT2 belts to drive the bevel gears; when both belts rotate in the same direction, the wrist performs a pitching motion, whereas when they rotate in opposite directions, the wrist executes a yawing motion. This configuration enhances mechanical efficiency and ensures precise control of wrist articulation.

ulation.

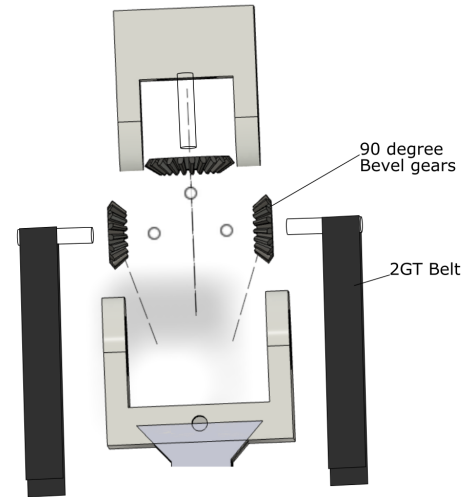


Figure 24: Wrist

4.7.2 Gripper

The gripper employs threaded mechanisms to convert rotational motion into linear movement, while parallel linkages are used to adjust the orientation of the linear displacement. This design ensures that the load cell remains consistently aligned, allowing for accurate force measurement and improved stability during grasping operations.

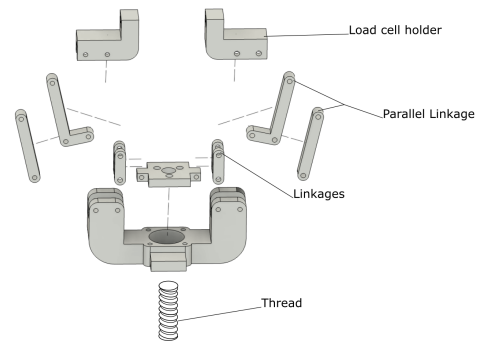


Figure 25: Gripper

5 Experimental Evaluation

Figure 26 and Figure 27 show the final version of our robot and its overall wiring layout. This design integrates all necessary components for our movement analysis experiments.

The robot is equipped with Mecanum wheels, which allow omnidirectional movement. Figure 28 presents a close-up of the Mecanum wheels used in our design. These wheels enable complex motion patterns, which are essential for our trajectory prediction experiments.

Figure 29 presents the 7-DoF robotic arm that provides improved flexibility and dexterity, allowing a wider range of motion and more precise manipulation of objects. Each degree of freedom contributes to independent movement in various directions, making the arm highly suitable for complex tasks requiring adaptability.

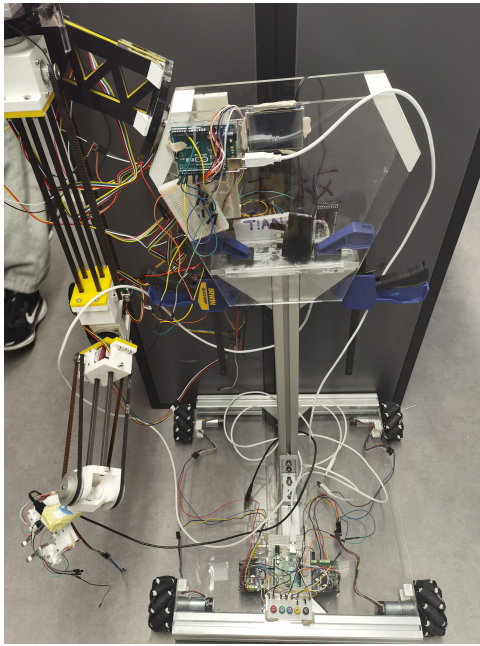


Figure 26: Final assembled robot

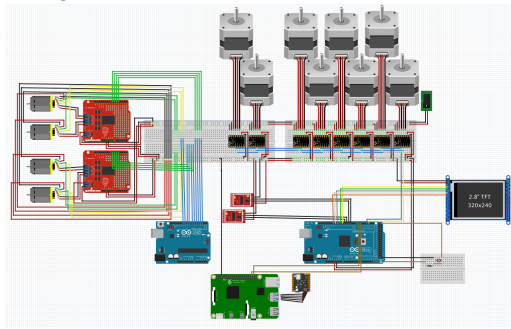


Figure 27: Overall wiring diagram

Figure 30 illustrates the wiring layout of the chassis and the robotic arm. The chassis utilizes an Arduino Uno connected to two L298P motor driver shields. Each shield controls two motors, enabling independent control of the four wheels. In addition, one motor is equipped with an encoder connected to the Arduino Uno, allowing closed-loop control of a single wheel.

1. Arduino UNO to L298P Motor Driver Wiring (Using External Power Source)

Arduino UNO Pin	L298P Pin	Description
D4	D4	Motor A Direction control
D5	D5	Motor A PWM control
D6	D6	Motor A PWM control
D7	D7	Motor A Direction control
D8	D4	Motor B Direction control
D9	D5	Motor B PWM control
D10	D6	Motor B PWM control
D11	D7	Motor B Direction control
GND	GND	Ground connection
– (external)	12V/VMS	External power positive
External GND	GND	External power ground

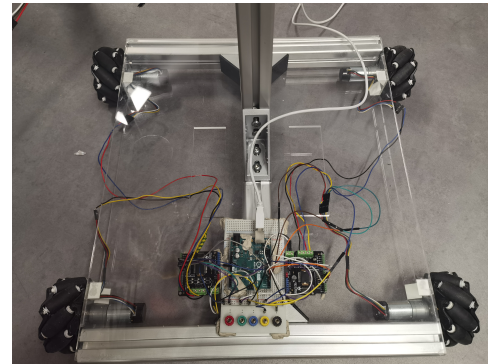


Figure 28: Mecanum wheels used in the chassis

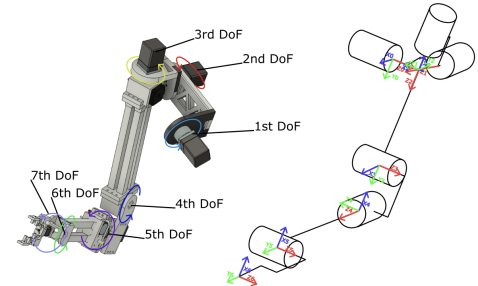


Figure 29: 7-DoF robotic arm

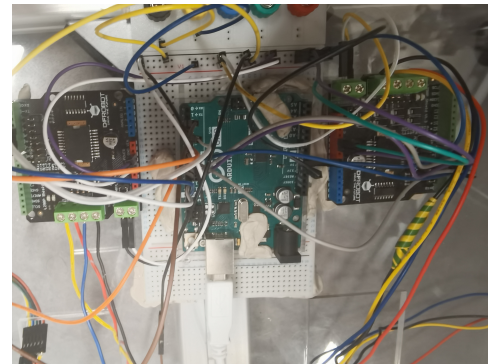


Figure 30: Chassis wiring layout

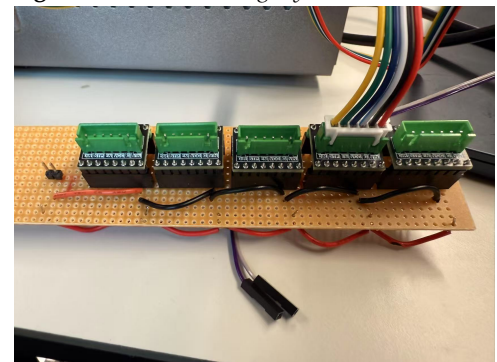


Figure 31: Robotic arm wiring

The motor equipped with an encoder can provide real-time feedback on the position and angle of the motor. When a specific angle is input, the wheel rotates accordingly, and the

encoder continuously reports the position in real time. Using PID control, oscillations are minimized and error is reduced, ensuring precise movement.

```
Current angle:0 Error: 0
New target angle set to: 1.00
Current angle:0.11 Error: -0.89
Current angle:0.37 Error: -0.63
Current angle:0.62 Error: -0.38
Current angle:0.89 Error: -0.11
Current angle:1.12 Error: 0.12
Current angle:1.05 Error: 0.05
Current angle:0.99 Error: -0.01
```

Figure 32: Motor with encoder for real-time feedback

2. L298P to Two Encoder Motors Wiring

Motor A&B:

L298P Pin	Motor Wire	Description
OUT1	M1+	Motor A&B positive
OUT2	M1	Motor A&B negative

Encoder Wire	Arduino UNO Pin	Description
A Channel	D2	Encoder channel A
B Channel	D3	Encoder channel B
VCC	5V	Encoder power
GND	GND	Encoder ground

In addition to the encoder, we also utilized other sensors, such as the Nicla Vision. The Nicla Vision camera was used to locate objects, capture images, and calculate their coordinates and rotation angles. Then, this information was transmitted to the gripper, enabling the robotic arm to move to the designated position and grasp the object precisely.

When using the gripper to grasp an object, feedback is necessary to control the gripping force. Otherwise, fragile objects may be crushed. To address this, we incorporated the HX711 load sensor. In addition, a display screen was placed in the center of the robot to monitor the readings of the load sensor in real time.

3. HX711 Load Cell and Arduino UNO Wiring

Load Cell to HX711:

Load Cell Wire Color	HX711 Pin	Description
Red (Exc+)	E+	Excitation positive
Black (Exc-)	E	Excitation negative
Green (Sig+)	A+	Signal positive
White (Sig-)	A	Signal negative

HX711 to Arduino UNO:

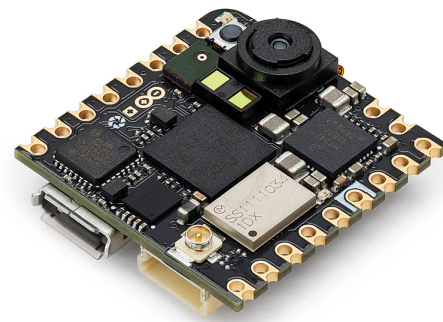


Figure 33: Nicla Vision sensor for object detection and positioning



Figure 34: Nicla Vision identifying and calculating object coordinates

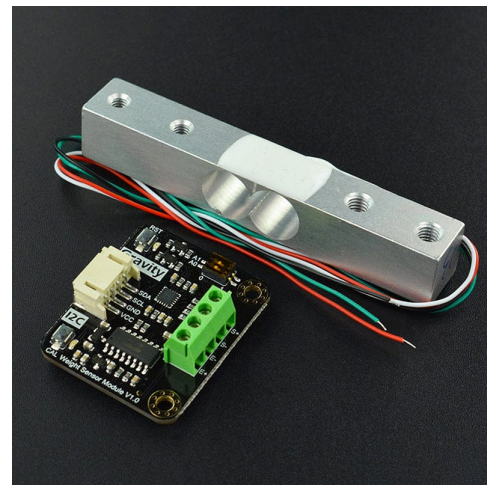


Figure 35: HX711 load sensor for force measurement

HX711 Pin	Arduino UNO Pin	Description
VDD	5V	Power supply
GND	GND	Ground
DT	D4	Data pin
SCK	D5	Clock pin

Regarding the display screen, we did not only show the load sensor readings but also the readings from an LDR sensor. We installed a GL55 Series Photoresistor on the robot. When

the surrounding environment darkens, the screen automatically dims to protect the eyes, and it brightens when the environment becomes brighter.



Figure 36: GL55 Series Photoresistor

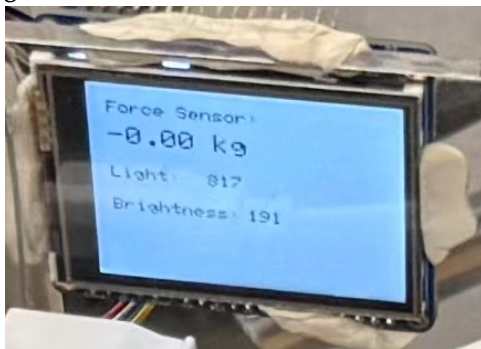


Figure 37: LDR and load sensor readings displayed on the screen

6 Discussion

The system currently relies on UART for communication with stepper motor drivers, but this approach requires multiple parallel cables, resulting in untidy wiring and limited scalability. Additionally, communication speeds are slow, especially as the number of devices increases. To address these issues, the CAN protocol can be implemented, offering higher communication speeds, better noise immunity, and supporting easier device expansion with fewer cables, thus improving scalability and data transfer efficiency.

The materials used in the construction of the robotic arm may deform under heavy loads, affecting precision. To enhance accuracy, materials such as aluminum alloy, titanium alloy, or carbon fiber should be considered, as they provide greater structural strength and reduce deformation under load, improving precision and stability during operations.

Currently, the system uses stepper motors, which have lower efficiency and can cause vibrations or overheating under high load. Replacing stepper motors with brushless motors equipped with encoders will improve efficiency, provide smoother motion control, and increase response speed, especially during high-speed or high-load tasks, thereby enhancing the overall performance of the robotic arm.

The Arduino Mega 2560 is used as the primary processing unit, but it lacks the processing power and flexibility necessary for handling complex real-time tasks. Replacing it with the STM32 microcontroller will significantly improve pro-

cessing power, I/O capabilities, and scalability, allowing the system to support more sensors, actuators, and communication modules, thus facilitating future growth and enhancing the robot's overall capabilities.

The current camera setup relies on single-image detection, which may result in inaccuracies due to object movement or processing delays. Switching to real-time video feeds will enable continuous object tracking, allowing the robotic arm to dynamically adjust its position and improve precision and responsiveness, especially in environments where objects are in motion.

The YOLO object detection model currently struggles to perform effectively under low light or in cluttered environments due to insufficient training data. Expanding the training dataset to include diverse lighting conditions, backgrounds, and object variations, along with using data augmentation techniques and multi-spectral sensing, will enhance the model's robustness and improve detection accuracy in challenging conditions.

Misalignment between the Nicla Vision camera's coordinate system and the robotic arm's coordinate system leads to errors in object positioning. To resolve this, the robotic arm's posture data should be repeatedly measured through a calibration board to determine the transformation matrix between the camera and robotic arm coordinate systems, ensuring more accurate conversions and better alignment.

The Nicla Vision camera currently lacks depth information, limiting the robot's ability to accurately calculate distances. The robotic arm can only reliably grip objects within a known range or distance. Adding depth-sensing capabilities, such as depth cameras, Time-of-Flight sensors, or LiDAR, will enable 3D object localization, improving the arm's ability to handle objects at varying distances and enhancing its performance in more dynamic environments.

The load sensor used in the system is not precise enough for detecting small mass variations, particularly when dealing with objects that weigh around 10 grams or less. By using a higher-precision load cell with sub-gram accuracy (e.g. 1-gram or smaller divisions), the robot will be able to handle smaller and more delicate objects with greater precision, improving its capability for tasks such as sorting, quality control, and fragile object handling.

7 Conclusion

The TianShu humanoid robotic system, with its Mecanum wheel base and advanced 7-DOF robotic arms, demonstrates significant potential for enhancing industrial automation. The integration of high-precision control systems, including Load sensor, along with a real-time Nicla Vision module and YOLO 11-OBb for object detection, allows the robot to operate with high flexibility, precision, and adaptability in dynamic environments. The robot's omnidirectional mobility, which allows it to navigate confined spaces, coupled with its precise motion and gripping controls, makes it a valuable innovation for industrial tasks such as assembly and defect detection. This capability also establishes a strong foundation for the development of more versatile and advanced humanoid robots in the future.

8 Appendix

We put all our code in a zip file, as we have different code for different part of the Robot.

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